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Steerable Liner While Drilling - A Proven Solution to Bridge Trouble Zones in Mature Assets

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Abstract

The Steerable Liner While Drilling (SLWD) technology has proven to be an effective solution for drilling through challenging formations such as rubble zones, unstable tar layers, and weak overburden. Today, this technology is also used to drill through different pressure zones in mature assets. This publication will describe the evolution of the Steerable Liner While Drilling system and outlines some of the case studies.

A Steerable Liner While Drilling system is a combination of advanced Rotary Steerable Drilling system and Liner Drilling allowing to drill complex wellbore trajectories. The liner is carried directly with the Drilling BHA. This concept mitigates the risk of hole stability issues and hole collapse between drilling and liner runs. After drilling to target depth, the pilot drilling BHA is disconnected from the liner which is left downhole. A typical application is where the section can be drilled easily but running the liner is problematic. The smearing, or plastering effect brings additional benefits in strengthening the bore wall.

Steerable Liner While Drilling technology is used for more than 15 years and has proven its functionality in various applications in new and mature fields. It has been used in different formations like sandstone and carbonates where unstable formations or depleted reservoirs lead to drilling- or liner running problems. The technology has evolved based on changing objectives and increasing requirements, becoming a strong and reliable solution to drill instable formations and to bridge different pressure zones in mature assets.

Originally developed to navigate trouble zones, SLWD technology has enabled faster wellbore construction in these applications and, in some cases, access to formations previously considered un-drillable using conventional drilling and completion technologies.

SLWD systems mitigate the risk of wellbore collapse or losing the wellbore before the liner is run.

Depleted reservoirs in mature assets becoming more and more the focus of operators globally. With the advent of second-generation BHA designs, SLWD continues to push the boundaries of what is technically feasible in complex drilling environments. The design has improved compared to first generation tools allowing longer and more reliable liner runs.

Introduction

Mature assets present a tremendous opportunity to boost hydrocarbon production and potentially lower cost per barrel (or MMscfd gas) produced. Currently, 20% of the global hydrocarbon production is associated with mature assets. This is poised to grow to 30% by 2030.

As the global energy consumption grows, all forms of energy will play a pivotal part in the mix. As per 2023 IEA World Oil energy outlook, oil will contribute at 33% and gas at 15% of the total energy consumption. Re-energizing existing oil and gas fields will be key driver to meeting this steep demand.

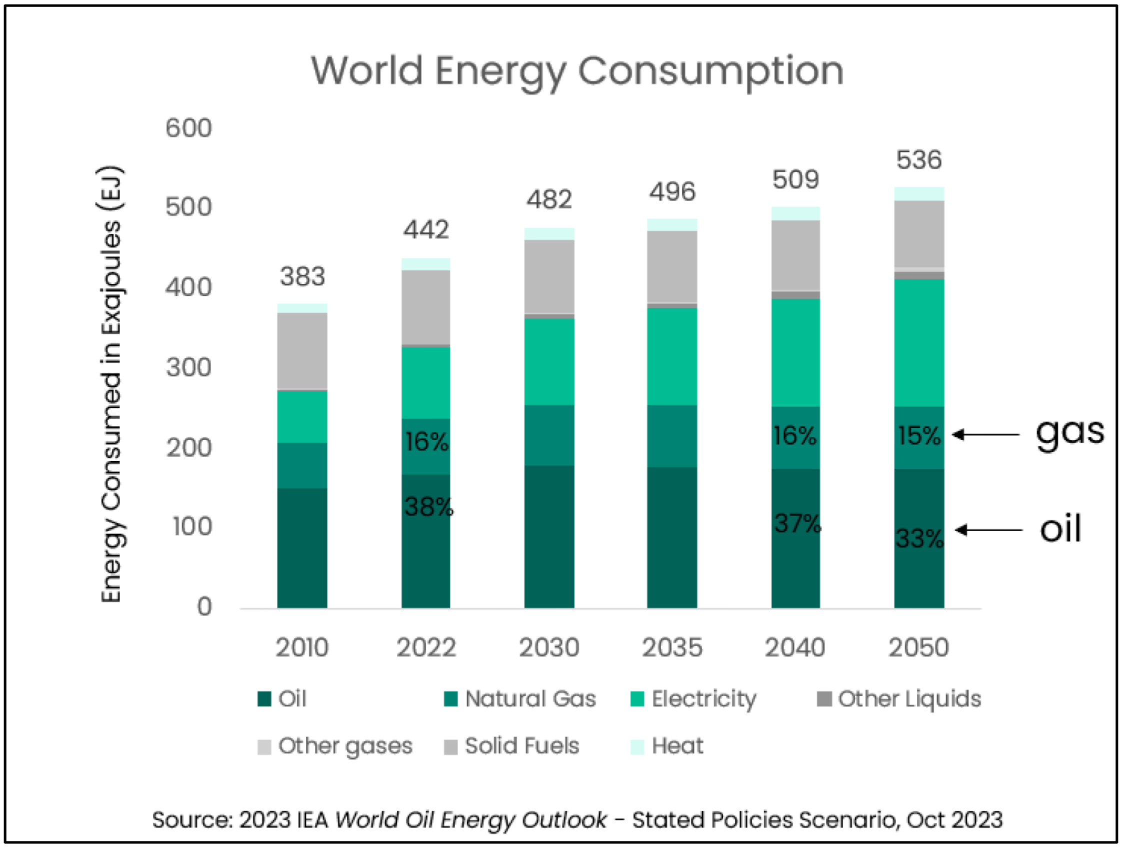


Figure 1—World Energy Consumption Forecast 2023 IEA World Oil Energy Outlook

Mature assets allow to produce short cycle barrel while reducing cost per barrel and carbon intensity by different means. These means could be by intervention, production optimization using chemical and mechanical methods, sand, water and gas management and lastly by unlocking bypassed reserves.

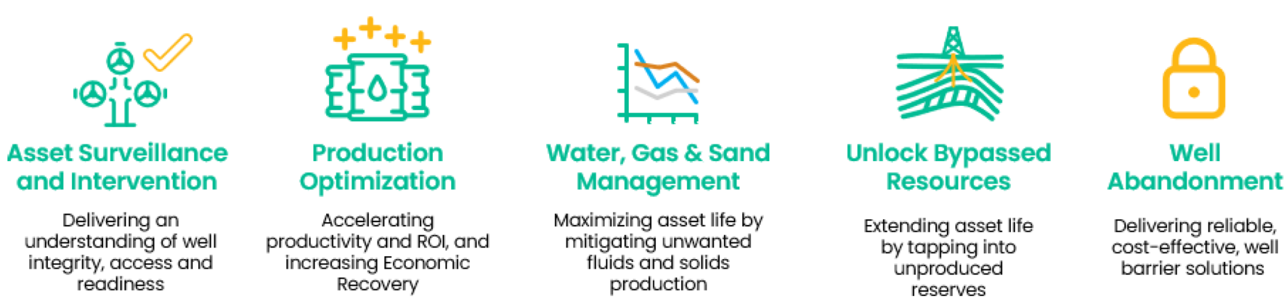


Figure 2—Mature Assets Solutions Themes for unlocking the potential of brownfields.

Bypassed reserves refer to hydrocarbon volumes left behind during the initial development of a reservoir, often due to unfavorable economics or the absence of suitable technical solutions at the time. These reserves may be located in compartmentalized zones, areas with poor sweep efficiency, or other geologically complex regions that prevent effective drainage by existing wells.

To access these stranded reserves, operators frequently turn to infill drilling or sidetracking. However, navigating through depleted reservoirs presents significant challenges. In many cases, the well path must pass through previously produced zones. Drilling through these zones with higher mud weights can lead to severe fluid losses, complicating well construction. Additionally, the pressure differential across the wellbore in depleted zones can be substantial, increasing the risk of wellbore instability and non-productive time.

These challenges directly impact operational costs. Given the late-life stage of many of these assets, operators are often under tight budget constraints, with limited tolerance for mud losses, stuck pipe, or other costly complications.

Steerable Liner While Drilling (SLWD) offers a cost-effective and technically robust solution for these scenarios. By enabling simultaneous drilling and casing, SLWD minimizes the exposure of unstable formations and mitigates fluid losses through the "smearing effect." This approach enhances wellbore stability and allows safe access to bypassed hydrocarbon zones that would otherwise be uneconomical or technically unfeasible to reach using conventional methods.

As mature assets continue to play a critical role in delivering secure, affordable, and sustainable energy, SLWD technology is proving essential in unlocking their remaining potential.

Casing and Liner Drilling systems

Today, casing and liner drilling systems are distinguished based on their design and application. These systems can be either non-steerable (Level 2) or steerable (Level 3). In casing drilling systems, the casing extends all the way to the surface. To operate these systems, surface equipment and the Blowout Preventer (BOP) must be modified to allow casing rotation from the surface. In contrast, liner drilling systems are conveyed on drill pipe. The drill string is rotated from the surface, requiring no modifications to the BOP, surface equipment, or connection procedures. Originally, both casing and liner drilling systems were non-steerable and used for short drilling intervals. Today, they remain suitable for applications where directional control is either unnecessary or not critical.

Steerable versions of both casing and liner drilling systems are now available, enabling the drilling of more complex well paths. A liner, being a shorter tubular conveyed on drill pipe, differs from casing, which is run to the surface. In addition to the benefits of using conventional surface systems and BOPs, liner drilling systems offer reduced torque and drag, allowing liners to be placed in deeper and longer sections compared to casing drilling.

Figure 3 provides an overview of existing casing and liner drilling systems. The Steerable Liner While Drilling (SLWD) system discussed in this paper is highlighted in red with bold lettering. This advanced SLWD system features a fully retrievable drilling Bottom Hole Assembly (BHA), including the latest Rotary Steerable System (RSS), Measurement While Drilling (MWD), and Logging While Drilling (LWD) technologies. It is classified as Level 4 in the hierarchy of casing and liner drilling systems.

The terms Steerable Liner While Drilling (SLWD) and Steerable Drilling Liner (SDL) are used interchangeably in industry literature.

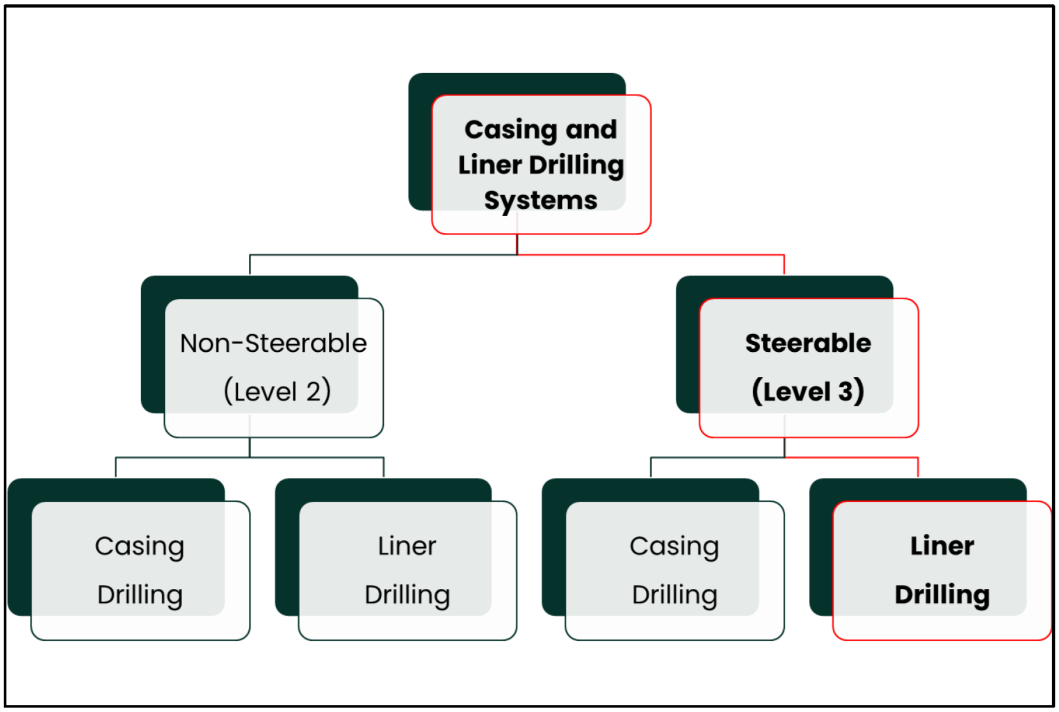


Figure 3—Overview of Casing and Liner Drilling systems

Benefits of Liner Drilling systems

The main advantage of Steerable Liner While Drilling (SLWD) systems is the integration of advanced technologies such as Rotary Steerable Systems (RSS), Measurement While Drilling (MWD), Logging While Drilling (LWD), and liner drilling. In this system, the liner is connected to the drilling Bottom Hole Assembly (BHA) and is run downhole simultaneously while drilling the well. When the BHA reaches total depth (TD), the liner is already at its target depth, eliminating the time and risk associated with running the liner separately. The fully retrievable BHA can include the latest RSS, MWD, and LWD tools, enabling precise geo-steering in complex three-dimensional wellbore geometries. The BHA is powered by a mud motor, which allows surface RPM to be minimized—just enough to overcome friction. The liner, which has a significantly larger outer diameter than conventional drill pipe, rotates slowly. This contributes to the "smearing" or "plastering" effect, which enhances the filter cake and strengthens the wellbore wall by pressing and compacting cuttings and formation debris against it. This effect helps mitigate lost circulation and improve wellbore stability. The phenomenon has been discussed in several industry publications. (e.g. Kumar 2013).

History of Liner while Drilling

Casing While Drilling (CWD) and Liner While Drilling (LWD) have a long and well-established history. As early as 1907, Reuben Carlton Baker introduced a cable drill casing shoe, marking one of the earliest innovations in this field. In the 1970s, a notable milestone was achieved with the drilling of 6,400 feet using 7-inch casing. During the 1990s, the use of these systems expanded, particularly in depleted reservoirs, driven by a growing focus on reducing flat time and improving operational efficiency.

Figure 4—cable drill casing shoes in the early 1900s (courtesy of Baker Hughes)

In 2005, a drillable drill bit known as EZCase was introduced and has since been deployed more than 400 times in sizes ranging from 4½" to 20". Despite its success, the casing and liner while drilling service faced two major limitations: it was not steerable and could not accommodate logging while drilling (LWD) tools.

Steerable Liner While Drilling Technology

The first generation of the Steerable Liner While Drilling (SLWD) system—also referred to as the Steerable Drilling Liner (SDL)—was developed between 2006 and 2010. Its purpose was to enable directional drilling and formation evaluation while simultaneously running the liner, thereby securing unstable or problematic formations as they were being drilled. In this system, the liner served as the outer string, while an inner string carried the pilot Bottom Hole Assembly (BHA), which provided steerability and logging capabilities. The pilot hole was enlarged using a swiveled reamer bit attached to a reamer shoe, which rotated at the same speed as the pilot BHA via a mud motor. Once total depth (TD) was reached, the pilot BHA could be released and retrieved from within the liner.

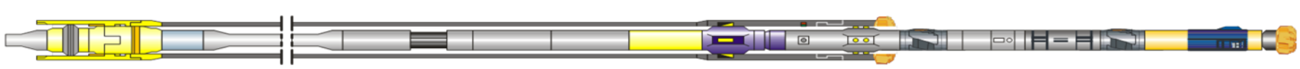


Figure 5—First generation Liner While Drilling system

The system was offered in 7-inch and 9⅝-inch liner sizes, combining equipment from Completions, Wellbore Interventions, and Drilling. While the 7-inch version was deployed in 16 jobs up to 2016, the 9⅝-inch version was only used once in 2009. Despite its technical success, there was no market demand for the larger size at the time, and the service was eventually discontinued. Although the 7-inch system had several successful runs, it was complex and faced reliability challenges. A key weakness was the hole-opening mechanism, which relied on a bit attached directly to the liner. A complicated procedure was required to connect a tool in the inner string to the inner diameter of the liner shoe to rotate the liner shoe bit. Additionally, if the bit became worn, the entire liner system had to be tripped out and laid down for replacement—though only two worn-bit incidents were recorded with the first-generation system.

The second generation of SLWD systems was developed between 2016 and 2018. It introduced a digitally controlled, downlink-activated underreamer integrated into the pilot BHA, eliminating several components required in the original design. This innovation allowed the main hole-opening tool to be replaced without pulling the liner. As shown in Figure 6, a bit is still retained at the liner shoe as a contingency in case a failure occurs near TD. Overall, the technical and operational complexity of the system was significantly reduced. The liner top equipment remains unchanged between the two generations. Figure 7 illustrates the differences between both systems.

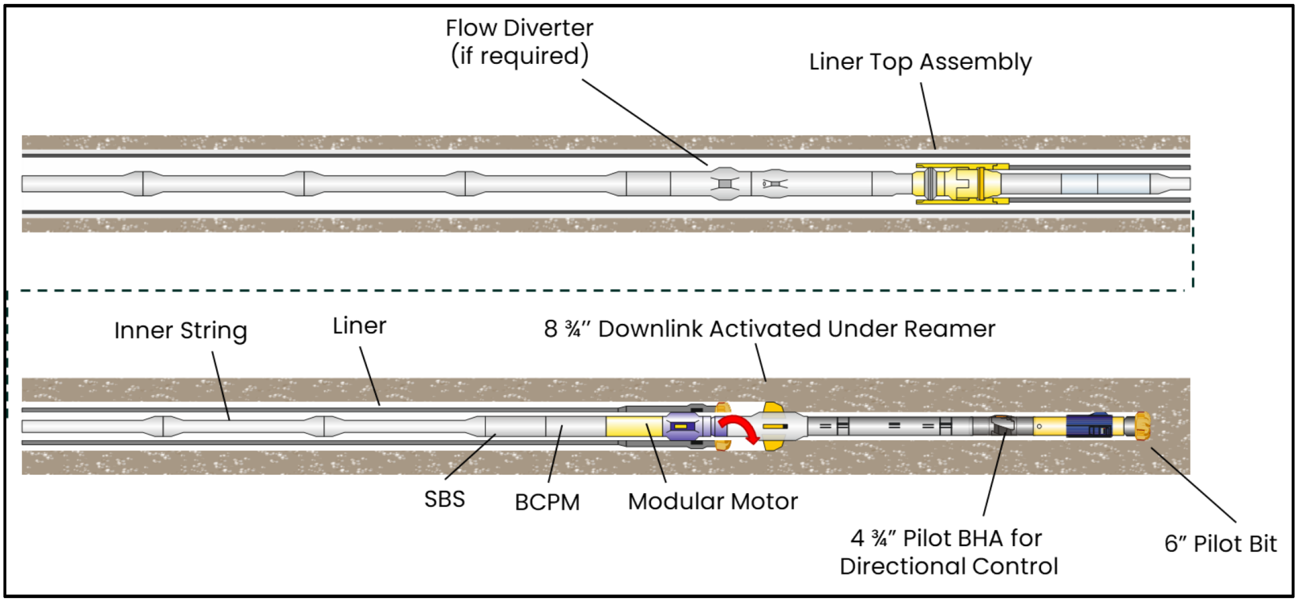


Figure 6—Second generation Liner While Drilling system (Bagga 2024)

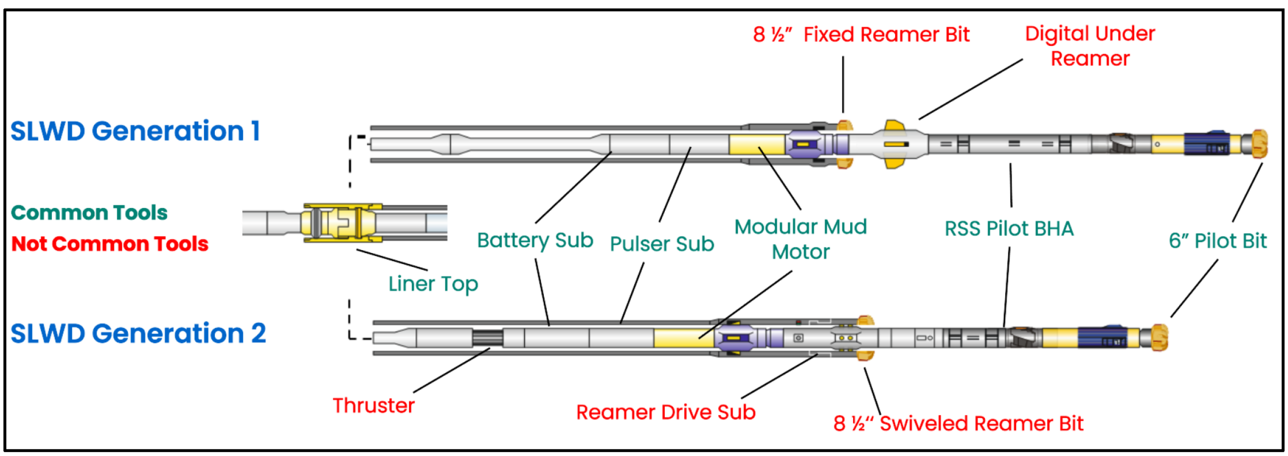


Figure 7—Difference between second generation (upper string) and first-generation (lower string) SLWD systems

As of today, the second generation of SLWD systems has just surpassed the number of deployments achieved by the original system. It demonstrates significantly higher footage per circulation hour, reflecting its improved reliability and operational efficiency compared to the first generation. The table below presents a comparison of key performance statistics between the two systems.

	First Generation SLWD System		Second Generation SLWD System
Liner size	9 5/8"	7"	7"
# of Jobs	1	16	19
Circulation hours	74	1177	1534
Drilling hours	32	540	1013
Distance Drilled (m)	180	4763	8369
Longest section (m)	180	1082	888
Liner Length (m)	1228	230-1560	168 – 1623

Figure 8—Difference between first generation and second generation SLWD systems

SLWD2 Improvements

After promising initial runs with the second-generation SLWD system in 2018, a noticeable increase in stall-out tendencies and mud motor failures was observed—despite using the same mud motor as the first-generation system. Through multiple iterations and investigations, the root cause was identified as the aggressiveness of the underreamer. Reamer blades are inherently aggressive by design to prevent weight stacking at the reamer, which is typically not problematic in conventional operations powered by a top drive. However, in SLWD applications using a mud motor, torque output is significantly limited—typically in the range of 2–3 kNm at operational differential pressure.

Additionally, weight transfer to the pilot BHA can be erratic due to the added friction from the liner, leading to sudden spikes in weight on bit (WOB) and corresponding torque surges. To address these issues, a redesign project was launched with two primary objectives:

1. Reduce aggressiveness to minimize motor stall-out risk
2. Improve lateral stability of the reamer

As a benchmark for the desired balance of aggressiveness, durability, and stability, the EZS305 8½" reamer bit from the first-generation SLWD system was selected. This bit had an excellent track record —across 16 jobs and 35 runs, it experienced no motor failures and only minimal stall-out incidents. However, the EZS305 is a five-bladed bit that resembles a coring bit more than a traditional reamer blade, presenting a challenge in adapting its properties to a three-bladed expandable reamer. The new design had to accommodate geometric constraints and stroke limitations while preserving the performance characteristics of the original bit.

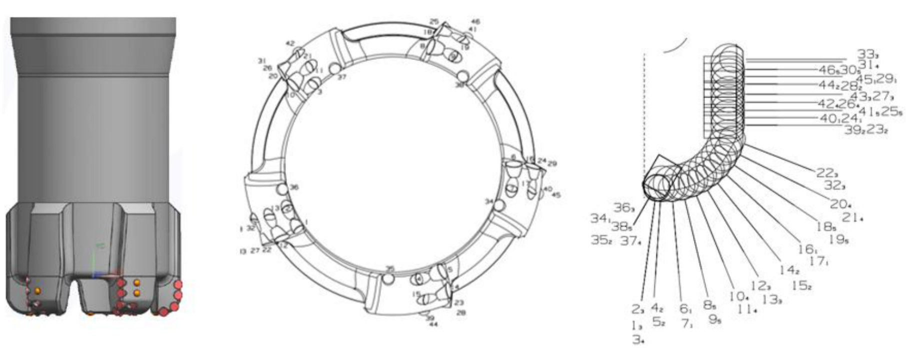
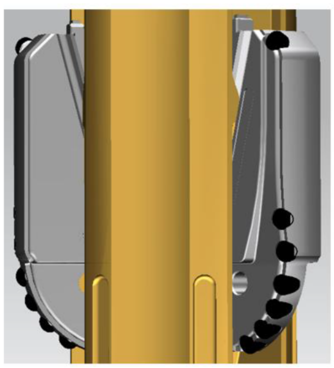


Figure 9—Design of the first generation SLWD bit

Drill bit fundamentals were applied to achieve the desired cutting behavior of the reamer blades:

- Shorten profile to improve lateral stability and reduce aggressiveness
- Use higher back rake angles to reduce aggressiveness
- Long gauge length improves lateral stability and encourages good borehole quality
- Apply premium cutter technology to maximize durability
- Sharper cutter chamfer improves efficiency and reduces torque requirement
- Use proprietary Tetrahedron cutter-rock interaction model to demonstrate changes in aggressiveness and durability

The result was a radical engineered design characterized by the short profile and long gauge length inspired by the EZS305



	Old Design	New Design
Profile	Long	Short
Backrake	20deg	25deg
Gauge Backrake	35deg	30deg
Gauge Length	2.0″	4.2″
Gauge Undersize	0.030″	0.050″
Cutter Size	13mm	13mm
Cutter Geometry	Stabilis X	0.016″ Planar
Ovoids	6 total @ 0.050″ under	None
StayTrue	7 total	None
Updrill	Yes	Yes

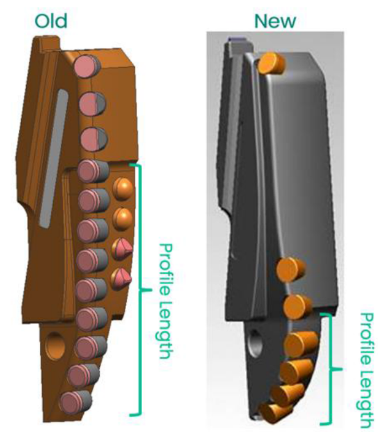


Figure 10—New and old reamer blade design

The Tetrahedron software played a crucial role in the reamer blade redesign process. Originally developed for standard drill bit modeling, the software was adapted to simulate reamer blade geometries as well. Although reamer blades had previously been modeled using other tools, Tetrahedron now enabled accurate comparative analysis across key parameters for the EZS305 bit, the original reamer blade design, and the new reamer blade concept.

One of the most revealing outputs was the Torque vs. Weight on Bit (WOB) plot, which illustrates the amount of torque generated for a given WOB. This chart provided valuable insight into the increased stall-out tendencies observed with the second-generation SLWD system. It clearly demonstrated how the more aggressive reamer blade design contributed to higher torque demands—exceeding the limited output capacity of the mud motor and leading to reliability issues.

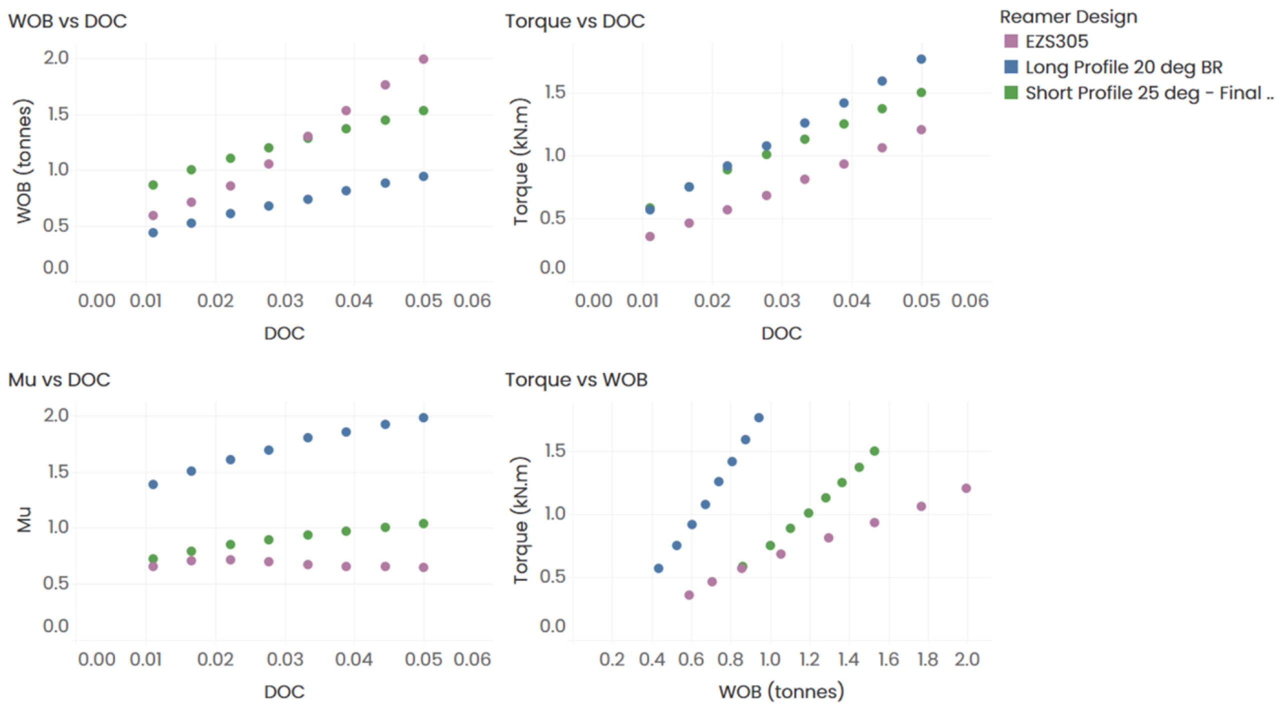


Figure 11—Simulated parameters for EZS305, old and new blade design

Due to the physical constraints and durability requirements of reamer blades, it was neither feasible nor expected to fully replicate the performance characteristics of the EZS305 bit. However, the new design achieved a 50% reduction in aggressiveness (Mu) compared to the previous blade configuration, while also enhancing lateral stability. Although this reduction in aggressiveness limits the maximum achievable rate of penetration (ROP), it was considered an acceptable trade-off to improve overall system reliability.

Field Test 1 – India

The first redesigned blade set was manufactured in 2021 but was not deployed until the second well of a four-well SLWD campaign in India, in December 2022. Although the first well (Well 1) was successfully drilled using the original blade design, it experienced frequent stall-outs, requiring constant monitoring of differential pressure. Since both wells were drilled in the same carbonate formation with similar well profiles, the performance data provides a reliable basis for evaluating the effectiveness of the design improvements.

Unfortunately, key parameters such as flow rate and torque were not made available by the contractor and could not be recorded. However, Figure 12 presents pump pressure as a function of measured depth for the two wells. Motor stall-outs are identifiable by drops in pump pressure, which occur when flow rate is reduced to pick off bottom and recover from a stall. In Well 1, this behavior was observed more than 14 times, whereas in Well 2 it occurred only once—and that was due to the string being run too quickly back to bottom.

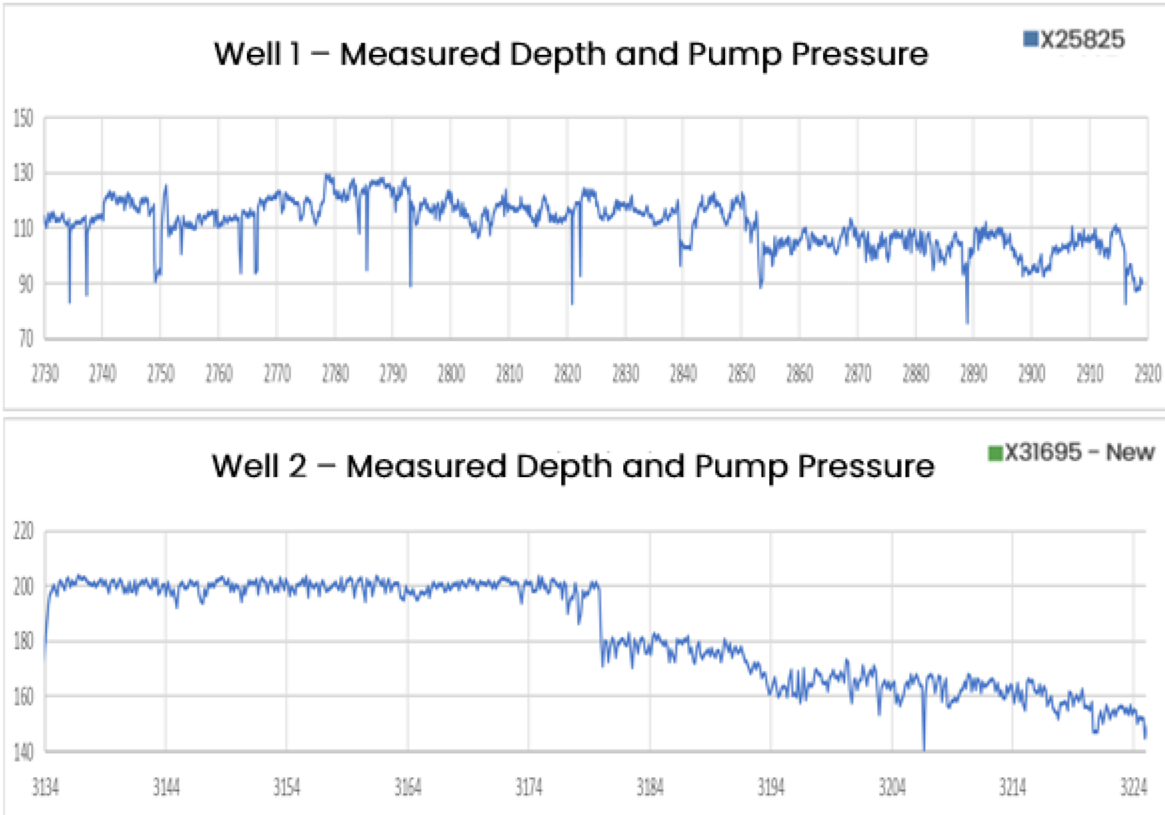


Figure 12—Pump pressure while drilling comparison between Well 1 and Well 2

The improved performance of the redesigned blade set was well received by both the crew and the customer. As a result, the same blade configuration was used in the remaining two wells of the campaign, delivering consistently good results. In total, the blade set drilled 574 meters over 61 drilling hours, with no significant wear observed on the blades.

Figure 13—New blade design after being downhole offshore India

Field Test 2– North Sea

In late 2023, the second-generation SLWD system was redeployed in the North Sea, following an unsuccessful attempt in the same field four years earlier. The previous run had been plagued by frequent stall-outs, ultimately resulting in mud motor failure. Given this history, expectations for the redesigned blade were high.

The section drilled measured 503 meters and carried a 554-meter liner. Geologically, the formation consisted of shale interspersed with paleosol layers, making it more prone to vibration than the carbonate formations encountered in the India campaign.

Both wells used identical BHAs, with one key difference: Well 1 employed a flow-split tool above the liner to enhance hole cleaning, while Well 2 did not require this solution. To better illustrate stall-out tendencies, flow rate and torque were plotted instead of pump pressure, as the latter was significantly affected by the flow-split tool in Well 1. In this setup, a drop in flow rate indicates a stall-out prevention or recovering from one.

Figure 14 compares a representative 142-meter interval from Well 1, drilled with the original reamer blade design, and Well 2, which used the new design. While Well 1 experienced five stall-out incidents during this interval, Well 2 had none. Additionally, torque readings in Well 1 were more erratic and spikier.

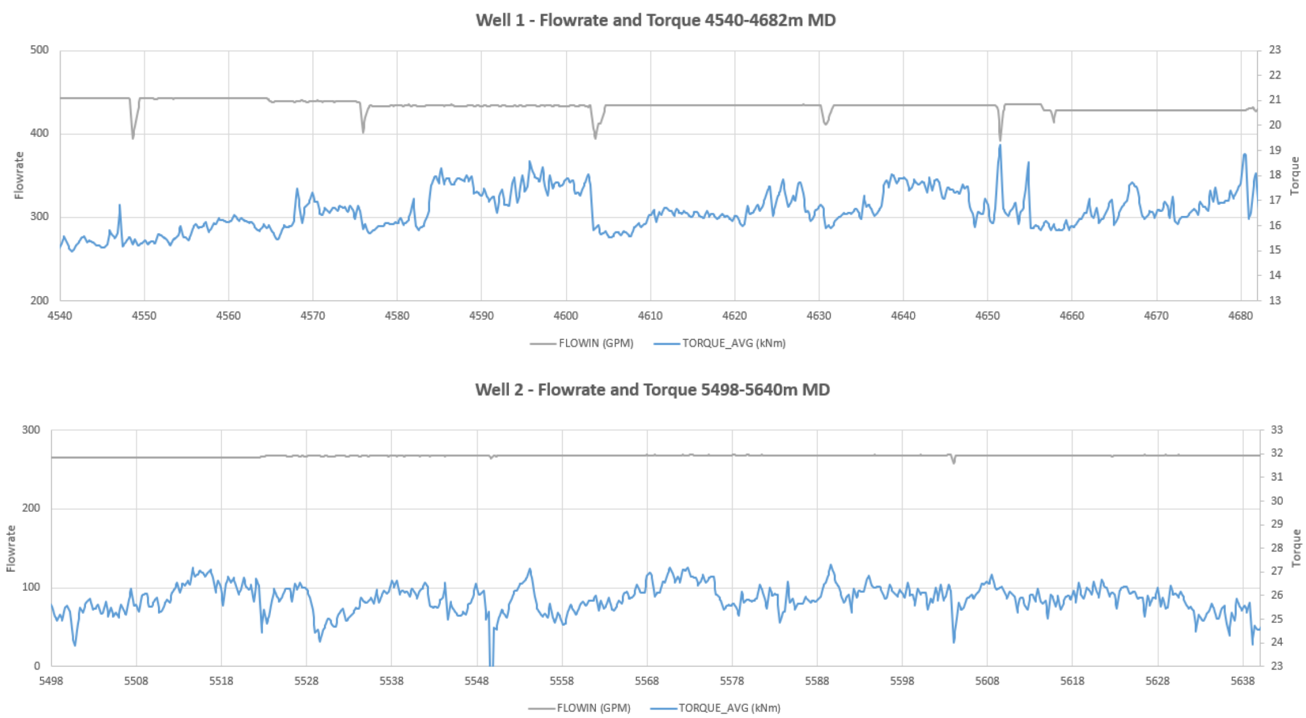


Figure 14—Flowrate and Torque comparison between Well 1 and Well 2

Field personnel noted that controlling differential pressure was much easier with the new design. Gradual pressure build-ups could be mitigated by slightly reducing weight on bit (WOB), rather than picking off bottom and lowering flow rate—a practice that was not feasible with the old design, where stall-outs occurred rapidly and unpredictably.

Vibration measurements from the underreamer were analyzed to assess the impact of the blade design. Both wells had comparable drilling durations—60 and 66 hours respectively—allowing for a meaningful comparison. To evaluate the influence of the new blade design, vibration levels were plotted as a percentage of total run time. Levels of 4 and above are considered potentially harmful if sustained over certain durations.

Figure 15 presents the vibration level distribution for both wells. The data shows that the duration of harmful vibration levels with the old blade design was approximately ten times greater than with the new design. This significant reduction is a strong indicator of the improved stability provided by the redesigned blades.

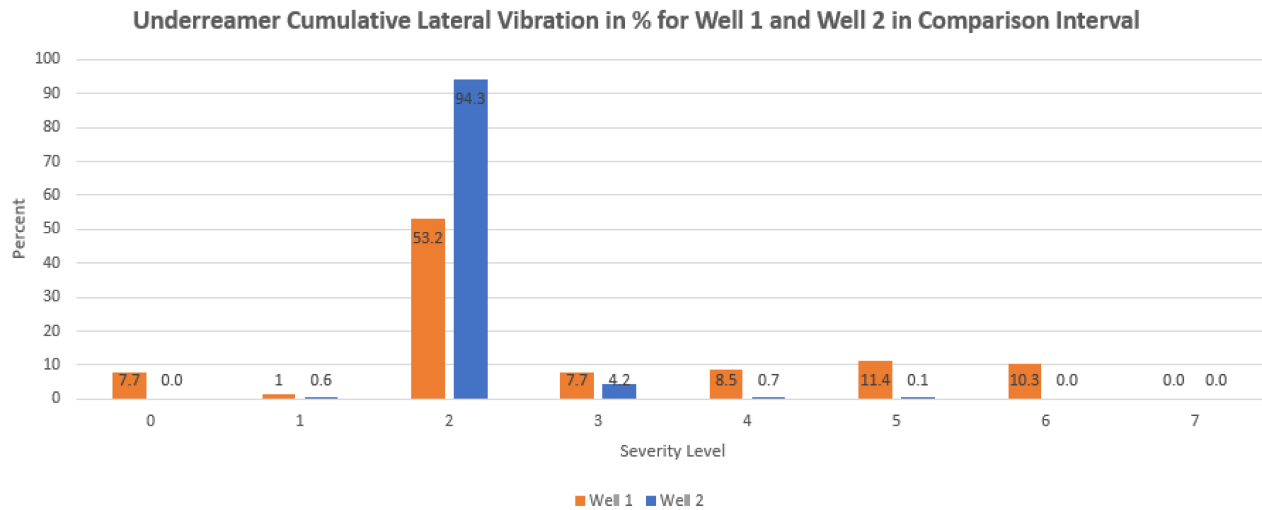


Figure 15—Vibration severity level comparison Well 1 and Well 2

Further evidence of the improved stability of the new blade design comes from the analysis of stick-slip levels, measured by the MWD sub located approximately 10 meters below the underreamer. While stick-slip is not considered as damaging to tools as lateral vibrations, it can reduce drilling efficiency and, in some cases, contribute to harmful lateral vibrations.

As shown in Figure 16, stick-slip levels were consistently lower with the new blade design, with the exception of level 7, where the new design recorded 0.2 additional hours. Overall, the data supports the conclusion that the redesigned blades contribute to a more stable and efficient drilling process.

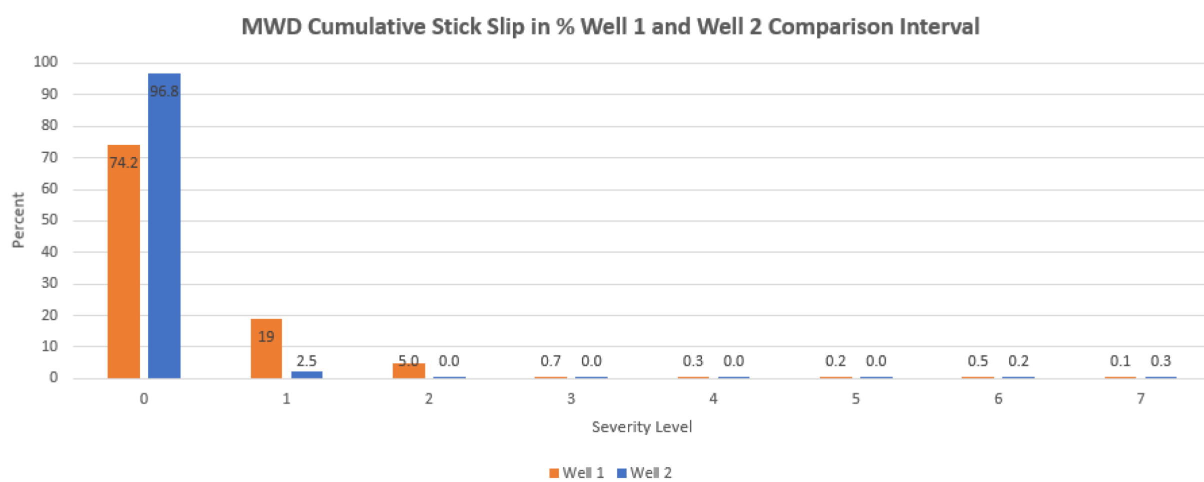


Figure 16—Stick Slip severity level comparison Well 1 and Well 2

Summary of improvements

The new reamer blade design for SLWD applications has demonstrated excellent performance in reducing motor stall-out tendencies, vibration, and stick-slip during underreamer operation. These improvements were achieved through an engineering-driven development process, supported by state-of-the-art simulation software that guided the design evolution. Field results have closely aligned with simulation predictions, confirming the effectiveness of the new design.

The outcome is a system that is easier to operate and offers a significantly higher likelihood of success. When combined with the second-generation modular motor and enhancements to the flow-split sub, the overall SLWD system has shown outstanding reliability in recent years, even in challenging drilling environments.

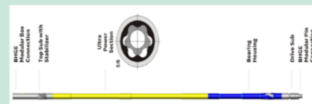
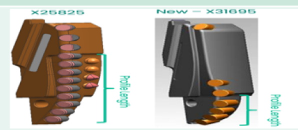
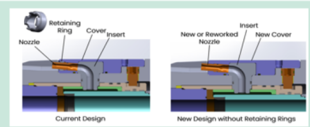
Tool	Issue description	Improvement Action	Year	Circ Hrs	Drill Hrs	Comment	Illustration	Issue Status
Motor	Old motor version was prone to chunk rubber prematurely during stallouts. Also incidents of sheared communications cable after stalls	Introduction of 4 ¾ MMTR2 with ultra rubber section and more robust drive train. Also less reactivity in oil based mud	2020	367	195	Good performance, not a single failure since implementation		**Solved**
GPE Series 6 Under Reamer	In certain formations the aggressiveness and lateral instability of the reamer blade caused frequent stallouts and high vibrations	New design based on reamer bit used in first SureTrak version has brought aggressiveness down. Increased gauge length for improved lateral stability	2021	200	110	Exceptional performance in both carbonate and shale formation		**Solved**
Flow Diverter	Nozzles lost (popped out)	New Nozzle design eliminates retaining ring which was root cause if lost nozzles	2020	269	133	No issues experienced since implementation		**Solved**
	Not able to close tool completely, small bypass still observed	Compensation piston upgrade						

Figure 17—SLWD2 Improvement overview

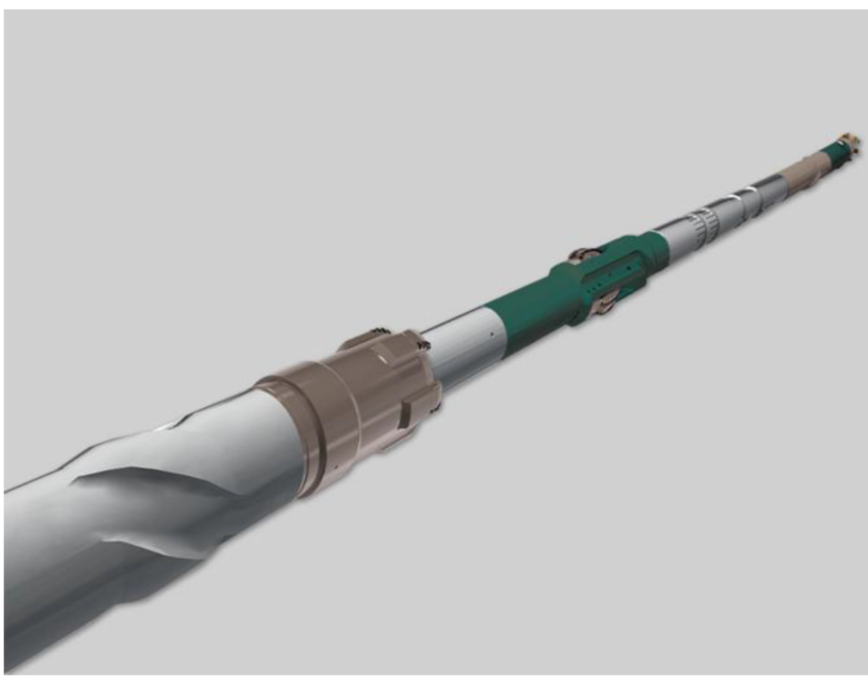


Figure 18—Second generation SLWD system

Alternate concepts - One-Trip System

Although the latest SLWD system enables drilling with a liner, cementing and setting the hanger-packer still require a separate run. To address this limitation, a One-Trip System (OTS) was developed, capable of performing these additional operations within the same drilling run. The OTS features a flapper valve integrated into the reamer shoe, a drilling-resistant hanger and packer, and a reconfigurable inner string that can shift positions to support the different phases of the operation—namely drilling, hanger setting, and cementing/packer setting.

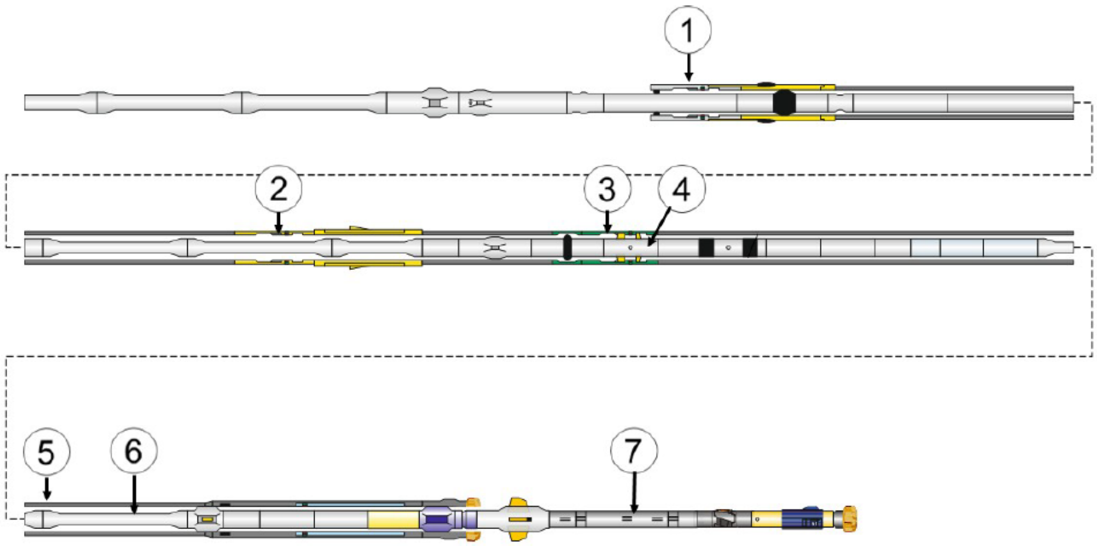


Figure 19—One-Trip system concept drawing

1	Cementing landing position and PBR sleeve	2	Reaming/ liner hanger setting position
3	Drilling landing position	4	Liner Drive Sub and Upper Assembly
5	Liner	6	Inner string
7	Pilot BHA		

Although the system was developed and successfully tested, changing market condition and elevated price prevented the technology from being launched commercially.

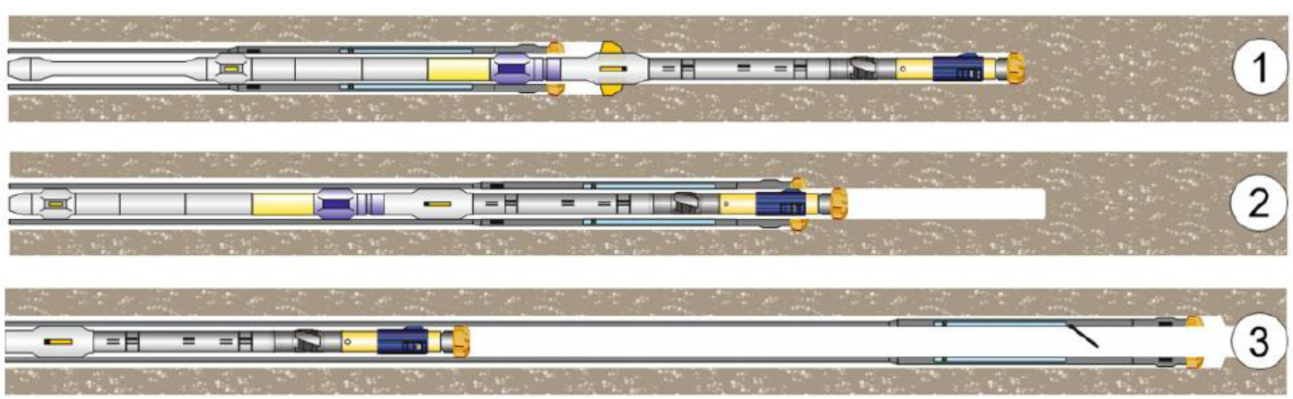


Figure 20—The three different operating positions of the One-Trip system

1	Drilling Position	2	Reaming/ liner hanger setting position
3	Cementing/ placker setting position		

Alternate concepts - Sacrificial Liner

The most recent innovation in SLWD has been the deployment of 4½" liners offshore Norway. In this configuration, the system drills a 6" pilot hole using a mud motor and a 4¾" Rotary Steerable System (RSS) BHA, with a crossover to the 4½" liner installed above the BHA. One limitation of this setup is that the BHA cannot be retrieved at total depth (TD), leading to the designation "sacrificial" for the system.

Despite this drawback, the design offers a key advantage: the liner can be equipped with pre-installed plugs and landing collars, enabling immediate cementing operations upon reaching TD. The liner top is run using the same tool as in larger SLWD systems, adapted to the smaller diameter.

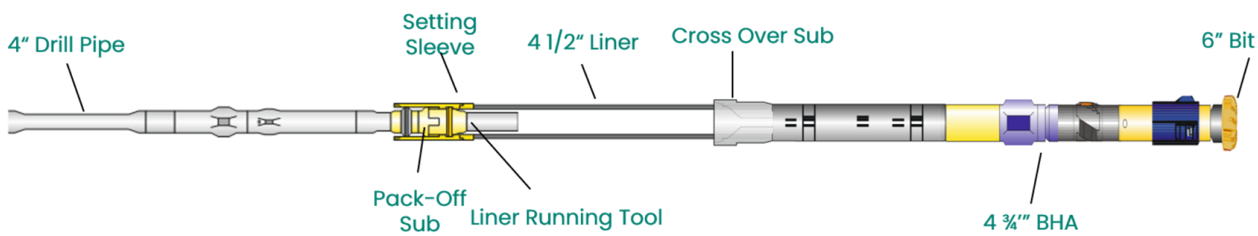


Figure 21—Sacrificial liner concept drawing

Case Studies

North Sea

Bossis et al. (2016) described significant challenges encountered in the Midgard field in the North Sea. The operator aimed to drill into the Garn formation to increase reserves and accelerate production. However, the unstable clay formations posed serious risks, particularly during liner running operations using conventional methods. Multiple attempts to run the liner after drilling failed due to formation collapse. Although the 8½" section was drilled without major issues, running the 7" liner proved impossible due to obstructions in the well. Several recovery attempts—including sidetracking and wellbore cleanouts—were unsuccessful, and ultimately, a drill string was lost in the hole, leading to temporary abandonment of the well.

To recover the well, a Steerable Liner While Drilling (SLWD) system—referred to as Steerable Drilling Liner (SDL)—was introduced, combining drilling and casing in a single run. Detailed planning, including simulations focused on equivalent circulating density (ECD), hole cleaning, torque, and drag, was critical for successful SLWD deployment. The sidetrack drilled with SDL showed excellent performance, with stable parameters and a rate of penetration comparable to conventional drilling. Despite high local doglegs, no significant drag was encountered, and hole cleaning was effective. Over one kilometer of the 8½" section was drilled with the liner in place, and the liner was successfully cemented after setting the hanger and packer. The SDL system enabled the economic recovery of a well that had previously failed using conventional methods.

It was concluded that due to time-dependent formation collapse, conventional drilling was not viable for this well. The SLWD system allowed simultaneous drilling and casing, ultimately enabling the operator to bring the well into production. This operation was carried out using the first-generation SLWD system, which was still considered a relatively new technology at the time and under continuous development. The second-generation SLWD system was launched two years later.

In another application, Noordoven et al. (2015) reported the use of SLWD from a semi-submersible rig in the Statfjord field. Drilling from a floating installation in rough weather conditions posed major challenges for both drilling and well control. Although offset wells had been drilled with minor difficulties, running the liner after pulling out the BHA often led to drill string pack-offs due to time delays, resulting in shallow liner or casing settings.

While casing and liner drilling were proven risk mitigation techniques in the area, SLWD was selected to drill more complex well trajectories. The well needed to be placed nearly parallel to the formation dip, with expected wellbore stability issues and varying pressure zones. A density-neutron measurement tool was added to the pilot BHA to support geo-steering. The well was successfully steered in response to geological features such as unexpected hard stringers.

A failure in the MWD component required tripping the pilot BHA to replace elements. After a ball drop, the inner string was released from the liner and pulled to surface. A new BHA was run, and circulation and rotation resumed after 54 hours. Drilling continued through multiple faulted layers and hard stringers, including five segments of unstable paleosol, without incident. The floater experienced heave up to 4.5 meters. Although the liner was set 135 meters short of the planned TD, the operation was considered a success.

The authors proposed technical improvements based on these experiences, which directly influenced the development of the second-generation SLWD system.

First Deployment of the Second-Generation SLWD System – Alaska Campaign

The first deployment of the second-generation Steerable Liner While Drilling (SLWD) system took place in 2018 during a six-well campaign in Alaska's Greater Mooses Tooth (GMT) project. The campaign concluded 2.5 months ahead of schedule, earning the supplier a recognition award from ConocoPhillips.

Grymalyuk et al. (2019) detail the early deployments of the enhanced 7" SLWD system in this project. The system was used to drill and simultaneously run liner in an 8¾" hole section through a highly unstable overburden, characterized by disturbed shale resulting from a prehistoric landslide and unknown bedding plane orientations. Conventional drilling methods had repeatedly failed in this area due to mechanical instability, particularly at high inclinations.

To overcome these challenges, the SLWD system was combined with Managed Pressure Drilling (MPD), enabling directional drilling while casing the wellbore in a single run. A 4¾" pilot BHA equipped with a digital, expandable underreamer enlarged the hole to 8¾", allowing the 7" liner to be deployed concurrently. Custom blade designs, MWD and LWD tools positioned below a motor, and the use of multiple cutting structures optimized performance. A new cementing method was also introduced to enhance reliability and provide contingency flexibility.

The system successfully drilled and cased the high-risk interval while acquiring valuable LWD data. The first two wells were completed in half the planned time, reducing drilling durations from 10 days to an average of just 2.5 days. No tool failures occurred, and a consistent rate of penetration (ROP) of 35–45 ft/hr was achieved. These results demonstrated the SLWD system's ability to reduce operational risk and cost while maintaining mechanical integrity and reservoir access.

Operator experience had previously shown that casing drilling could mitigate mud losses and hole collapse via the "smear effect," but torque limitations and fatigue risks made it unsuitable for the GMT project. The first-generation SLWD system addressed torque and directional control but suffered from low ROP. The second-generation system introduced a digitally controlled underreamer with expandable blades, significantly improving stability and performance. This underreamer could be retracted for blade replacement without pulling the liner.

The system also featured a rotary steerable device, MWD/LWD tools, and a smart circulation sub for optimized flow and hole cleaning. Real-time measurements ensured dynamic stability, and surface commands enabled unlimited tool activations. After reaching total depth, the inner string was retrieved, and cementing was performed using a mechanically set liner hanger and packer. The new cementing method offered simplicity and flexibility, reducing the risk of hydraulic failures and enabling effective cement placement even in partial pack-off scenarios.

The primary objective of the project was early and reliable production delivery—an outcome at high risk due to the unstable overburden encountered during exploration. SLWD technology proved essential, particularly in the final well, where seismic uncertainty led to premature entry into the problematic zone, resulting in stuck pipe and lost circulation. After cleaning out the casing shoe, the SLWD system was deployed and successfully drilled the remaining section to TD, confirming its ability to recover difficult hole sections and maintain well integrity.

The system's performance exceeded expectations, validating its necessity for the project. Its success was attributed to both technological innovation and strong execution by the operational team. Ultimately, the project achieved production goals that would have been unattainable using conventional methods.

India

Between 2022 and 2024, a four-well pilot campaign using the second-generation Steerable Liner While Drilling (SLWD) system was conducted offshore Western India. This marked the first deployment of SLWD technology in Asia, targeting carbonate formations known for severe loss circulation and liner deployment challenges when drilled conventionally. All wells featured similar profiles, with tangents around 77° and section lengths of approximately 200 meters. Liner lengths ranged from 400 to 631 meters to provide a safety margin for cement displacement over the liner top.

Well 1. The first well was drilled with partial losses and experienced frequent stall-outs, resulting in a low average rate of penetration (ROP). Cementing and setting of the liner hanger and packer were completed in two additional runs. Despite the challenges, the cement bond log (CBL) showed acceptable cement quality.

Well 2. The second well marked the first use of the newly designed reamer blades, which delivered excellent performance—no stall-outs occurred during the run. However, after 100 meters of drilling, total losses were encountered, requiring seawater to be pumped. This led to a significant torque increase, and after 161 meters, the top drive torque limit was reached, forcing TD to be cut short by about 70 meters. Despite this, two of the three geological targets were reached. Cementing was performed using the same method as in Well 1, but due to total losses, the CBL indicated poor cement coverage, necessitating a remedial squeeze job.

Well 3. This well experienced only moderate losses and further highlighted the efficiency of the new reamer blade design, achieving an average ROP of 13.3 m/hr—more than double that of the previous wells (5.6 m/hr). A flow diverter was used to enhance hole cleaning before reaching TD. Cementing was again performed under loss conditions, requiring another squeeze job.

Well 4. Initially, total losses were observed, but they later stabilized at 220 bbl/hr. This well involved more directional steering, with doglegs up to 2.5°/30m. Planned TD was reached, and the liner was disconnected without issue. Unlike the previous wells, completions were performed in a single run. Despite flawless execution, cementing was again performed under loss conditions, requiring a remedial squeeze job.

As summarized by Bagga (2023), this campaign made history as the first successful SLWD deployment in Asia. The operation was made possible through close collaboration between the National Oil Company and five product lines from Baker Hughes. The campaign demonstrated that SLWD technology could effectively address severe loss circulation challenges while drilling, delivering four wells that would have been extremely difficult and time-consuming to complete using conventional methods.

Sacrificial liner

The sacrificial liner concept has been deployed twice offshore Norway, targeting unstable paleosol and coal formations. This application of Steerable Liner While Drilling (SLWD) technology enables access to remaining oil reserves that are unreachable using conventional drilling methods. Unlike Level 3 and 4 SLWD systems, the sacrificial liner setup does not involve hole enlargement, simplifying the Bottom Hole Assembly (BHA) configuration.

In this system, the BHA is connected to the bottom of the liner via a crossover, featuring a premium liner connection box up and a BHA connection pin down. A 6" pilot hole is drilled using a Rotary Steerable

System (RSS) BHA, which pulls a 4½" liner behind it. At the top of the liner, a conventional liner running tool is used, which can be disconnected via a ball drop mechanism.

In the most recent deployment, a 1,258-meter liner was successfully installed. The coal and paleosol layers were drilled without issue. Pack-offs were encountered when entering sand formations later in the section, but these were managed effectively by carefully controlling flow rate and penetration rate. After 858 meters of drilling, the rig experienced a 28-hour generator outage. During this downtime, slow back-reaming was performed using cement pumps at 250 LPM. Drilling resumed for an additional 70 meters before further generator issues prompted the operator to call total depth (TD) early. The liner running tool was then disconnected, leaving the liner and BHA in place.

Despite the early termination, the run was considered a success. A total of 928 meters of partially unstable formation was drilled and cased using the sacrificial liner, achieving an average ROP of 11.7 m/hr. Although cementing was planned, it could not be carried out due to the generator problems.

Conclusion

Over the past 15 years, the Steerable Liner While Drilling (SLWD) system has evolved into a commercially viable and field-proven technology. It offers a cost-effective solution for drilling through problematic zones in mature assets, where conventional methods often fall short. By improving wellbore stability and mitigating mud losses through the "smearing effect," SLWD enhances both safety and well control. In many cases, it has enabled access to hydrocarbon reserves that were previously unreachable using conventional drilling techniques.

The 7-inch SLWD system is the most widely used configuration globally. However, certain applications require different hole sizes that are incompatible with retrievable SLWD technology. In such cases, non-retrievable, sacrificial SLWD systems are deployed, with liner sizes as small as 4½ inches. These systems are tailored for specific challenges, such as unstable formations or narrow wellbores.

The technology was continuously improved over two decades. Today, the standard SLWD technology is very robust and reliable. Ongoing enhancements and adaptations are preparing it to meet the demands of a fast-changing energy landscape.

A One-Trip-System which allows drilling, running liner, setting hanger and packer and finally cementing in just one run was developed, but did not prove its suitability for the market. This system became overly complex and costly. Therefore, running a regular SLWD system with additional one or two trips for cementing and setting liner hanger and packer is more economical.

As operators continue to sidetrack aging wells and pursue deeper reservoirs, SLWD offers a practical solution for navigating depleted zones and unstable formations. SLWD provides a solution to access deeper hydrocarbon reservoirs while drilling through depleted zones safely. With depletion of mature assets, there is a noteworthy change pore pressure, fracture gradient and collapse pressure regimes within the formation. SLWD also provides a solution to drill through unstable formation. In addition, due to depletion, loss zones with high mud weight maybe a challenge to both drill and run casing/liner which can be combated with SLWD solution. As mature assets take center stage in delivering sustainable, affordable, and secure energy, bypassed hydrocarbons remain a critical target—and SLWD is proving to be a key enabler in unlocking their potential.

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